

MTH401 FINAL TERM SOLVED PAPER

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1. The wronskin of the function $y_c = c_1 + c_2 \cos x + c_3 \sin x$ is

lineraly independent

lineraly dependent

both of the above

none of the above

2. If the auxiliary equation has two roots $m_1 = 2$ and $m_2 = 2$ then the general solution is

$$y = c_1 e^{mx} + c_2 e^{mx}$$

$$y = (c_1 + c_2 x) e^{mx}$$

$$y = c_1 e^{\alpha x} + c_2 e^{\alpha x}$$

$$y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x$$

Integarting factor if the D.E $\frac{dy}{dx} + \frac{y}{x \ln x} = \frac{3x^2}{\ln x}$ is

$$\ln x$$

$$x \ln x$$

$$\frac{1}{\ln x}$$

$$\frac{1}{x}$$

The integrating factor of the D.E $\frac{dy}{dx} + y \ln y = y e^x$ is

$$e^x$$

$$e^y$$

$$e^{\frac{1}{x}}$$

$$e^{\frac{x}{y}}$$

The interval of convergence for the function $\sec x$ is

$$(-\pi, +\pi)$$

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

correct answer

$$\left(\frac{\pi}{2}, \pi\right)$$

None of them

$$\frac{dy}{dx} - 2xy = 0$$

The solution of the linear first order differential equation is

$$y = e^{x^2}$$

Correct answer

$$y = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

$$y = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

Both $y = e^{x^2}$ &

None of them

- Which of the rule in matrices under multiplication does not hold true?

Commutative law

Associative law

Identity law
None of them

- Any two matrices A and B are said to be equal if and only if they have the same orders and the corresponding elements of the two matrices are equal. Then which of the following is true:

A=B

AB=BA

A≠B

None of the

given

$$\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \end{pmatrix} \sin t$$

The given system without the use of matrices is

$$\frac{dx}{dt} = 3x - 7y + 4 \sin 2t; \quad \frac{dy}{dt} = x + y + 8 \cos 2t$$

$$\frac{dx}{dt} = 3x - 7y + 4 \sin t; \quad \frac{dy}{dt} = x + y + 8 \cos t$$

$$\frac{dx}{dt} = 3x - 7y + 4 \sin t; \quad \frac{dy}{dt} = x + y + 8 \sin t$$

correct

Suppose that $\{X_1, X_2, X_3 \dots X_n\}$ is a set of n solutions vectors on an interval I , of a homogeneous system $X' = AX$. The set is said to be a fundamental set of solutions of the system on the interval I if the solution vectors are

Linearly dependent

Linearly independent

Correct answer

Homogeneous

None of them

$$A = \begin{bmatrix} 3 & -18 \\ 2 & -9 \end{bmatrix}$$

The matrix has an eigen value of multiplicity

1

2

3

4

If L denote the linear differential operators with constant coefficients, then $L_1 L_4 - L_2 L_3$ represents the

$$\begin{vmatrix} L_1 & L_2 \\ L_4 & L_3 \end{vmatrix}$$

$$\begin{vmatrix} L_1 & L_3 \\ L_4 & L_2 \end{vmatrix}$$

$$\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix}$$

Correct answer

None of them

Any linear differential equation of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 x \frac{dy}{dx} + a_0 y = g(x) \text{ where } a_0, a_1, a_2, \dots, a_n \text{ are constants.}$$

is called

Homogeneous equation

Polar equation

Equi-dimensionl equation or Cauchy Euler correct answer

None of them

$$A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$$

For eigen values $\lambda = 5, 5$ of a matrix, there exists eigen vectors.

infinite

one

two

three

Let λ be an eigenvalue of a none zero square matrix A. Then the equation

$$\det(A - \lambda I) = 0$$

is called

Trivial equation

Characteristics equation

Non-trivial equation correct answer

None of them

chracterstic

equotaion

For $y \sin^2 x - y^2 \cos x = c$ where $y(0) = 3$; the value of c is

9

-9

10

-10

A differential equation $M(x, y) dx + N(x, y) dy = 0$ is said to be an exact if -----.

$$\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \text{Correct answer}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$$

- What is the nature of the roots of the following D.E

$$x^2 \frac{d^2 y}{dx^2} - 5x \frac{dy}{dx} + 8y = 0$$

Repeated root

Real and distinct

Complex or imaginary

None of the above

- If $E(t)=0$, $R \neq 0$ Electric vibration of the circuit is called_____

Free damped oscillation

Un- damped oscillation

Over damped oscillation

None of the given

- If $E(t)=0$, $R = 0$ Electric vibration of the circuit is called_____

Free damped oscillation

Un- damped oscillation

Over damped oscillation

None of the given

- The reactance and impedance both are measured in_____

Ohm

Newton

Volt

None

- Eigen value of a matrix $\begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$

5, 5

10, 5

25, 5

None

If a matrix has 3 rows and 1columns then the given matrix is called_____

Column matrix

Row matrix

Rectangular matrix

Square matrix

- The matrix $\begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$ is: $6-6=0$

Scalar matrix

Singular Matrix $6-6=0$

Non Singular matrix

None

The solution of the differential equation $\frac{dy}{dx} \tan x$ is_____

$y = \ln|\sec x| + c$

Sec x

Ln sec x+c

None of the given

Factorial of n =

(n+1)

n!

n+n=1

none of the above

Q1:Write down the system of differential equations

(5marks)

$$\frac{dx}{dt} = 6x + y + 6t, \quad \frac{dy}{dt} = 4x + 3y - 10t + 4$$

in form of $X' = AX + F(t)$

Solution:

$$X' = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} X + \begin{pmatrix} 6t \\ -10t + 4 \end{pmatrix}$$

Write name of 5 kinds of functions which can be differentiated

(5marks)

Solution:

Linear function

Non-linear function

Homogeneous function

Non homogeneous function

Complementary function

Bessel function

$$\frac{dy}{dt} = x, \frac{dx}{dt} = y$$

- Solve the system of differential equations by systematic elimination.
(5Marks)

Solution:

$$\frac{dy}{dt} = x \Rightarrow Dy - x = 0 \quad \dots\dots(i)$$

$$\frac{dx}{dt} = y \Rightarrow -y + Dx = 0 \quad \dots\dots(ii)$$

Operate (ii) by D , we get

$$-Dy + D^2x = 0 \quad \dots\dots(iii)$$

Add (i) and (iii), we get

$$Dy - x = 0$$

$$\underline{-Dy + D^2x = 0}$$

$$D^2x - x = 0$$

$$(D^2 - 1)x = 0$$

Auxiliary equation is $m^2 - 1 = 0$

$$m = \pm 1$$

$$\boxed{x(t) = c_1 e^t + c_2 e^{-t}}$$

Put this in (i), we get

$$Dy - [c_1 e^t + c_2 e^{-t}] = 0$$

$$Dy = c_1 e^t + c_2 e^{-t}$$

Integrate both sides, we get

$$\boxed{y(t) = c_1 e^t - c_2 e^{-t}}$$

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